Kaitlynn T. Pineda

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RESEARCH OVERVIEW

My research focuses on how LLMs can support natural small talk in physically collaborative robots. I design, develop, and deploy autonomous robotic systems that integrate social dialogue into task-based scenarios to foster engagement and rapport in human-robot/agent teams. My work spans lab-based human-subject studies and real-world field deployments.

EDUCATION

Johns Hopkins University, Baltimore, MD 08/2021 – Present
PhD Candidate *in Computer Science*
Advisors: Chien-Ming Huang and Gregory D. Hager

Johns Hopkins University, Baltimore, MD 05/2024
MSE *in Computer Science*

Yale University, New Haven, CT 08/2017 – 05/2021
Bachelor of Science *in Electrical Engineering and Computer Science,* ***Certificate*** *in Spanish*
Advisor: Brian Scassellati

RESEARCH EXPERIENCE

Johns Hopkins University, Baltimore, MD 08/2021 – Present
Graduate Research Assistant | Intuitive Computing Lab & Laboratory for Computation Sensing and Robotics

- Create LLM-driven robotic systems that integrate small talk into physical tasks, conducting lab studies and "in-the-wild" system deployments to evaluate engagement, rapport, and teamwork in human-robot collaborative settings
- Apply multimodal analysis methods including mixed-effects modeling, behavioral coding, and facial/gaze signal processing, and mixed qualitative-quantitative empirical evaluations

Yale University, New Haven, CT 05/2018 – 05/2021
Undergraduate Research Assistant | *Social Robotics Lab*

- Developed Unity-based interactive game interface for user studies on trust & fairness in human-robot collaboration
- Conducted behavioral analysis of children with ASD using a long-term, in-home socially assistive robot

Université catholique de Louvain, Louvain-la-Neuve, Belgium 05/2019 – 07/2019
Research Assistant | *ICTEAM*

- Applied 2D and 3D U-Net models for biomedical image segmentation of mice entheses and integrated autoencoder based priors to enhance morphological accuracy

PROFESSIONAL EMPLOYMENT

Nissan, Santa Clara, CA
Artificial Intelligence Intern | *NATC-SV* 06/2025 – 08/2025

- Conducted R&D on LLM-driven conversational agents for human-vehicle interactions; contributed to a cross-team demo for Nissan's Global CEO, Ivan Espinosa, showcasing interactive AI experiences for autonomous vehicles
- Designed agentic workflows with RAG and function calling for conversational systems using speech & vehicle data
- Prototyped and deployed on-device conversational agents powered by local LLMs for real-world vehicle contexts

Meta, Menlo Park, CA
Software Engineering Intern | *Meta Quest (Oculus)* 06/2021 – 08/2021

- Created a synthetic IMU trajectory generation pipeline within Meta Reality Labs, producing artificial kinematic headset data from upper-body and head animation key-point sequences
- Utilized Monte Carlo sampling for spatial optimization in virtual environments and used MATLAB to visualize and validate 3D trajectory alignment algorithms

Software Engineering Intern | *FAIAR* 06/2020 – 08/2020

- On the AI Applied Research – Conversational AI team working on dialog policy for Meta Ray-Ban Glasses
- Built Android & web testing platforms to streamline photo/video transfer pipelines from devices to cloud services

PERSONAL AWARDS

[11] **Professor Joel Dean Excellence in Teaching Award**, Computer Science Department 2025

- Awarded annually at JHU in recognition of extraordinary performance in teaching undergraduates.

[10] **Graduate Student Employee of the Year (nominee)**, Johns Hopkins University 2025

- Awarded annually for a student's significant contribution to the university through their employment position

[9] **Best Paper Honorable Mention, ACM/IEEE International Conference on Human-Robot Interaction** 2025

- *Impacts of Robot Small Talk on Task, Rapport, and Interaction Dynamics in Human-Robot Collaboration.*

[8] **IEEE Robotics and Automation Society (RAS) Travel Support Award** 2025

- Student Travel Support for the HRI 2025 Conference

[7] **Robotics Science and Systems (RSS) Inclusion Fellow** 2022

- Fellowship providing full conference registration and travel support, mentorship, and cohort programming

[6] **Computer Science Departmental Fellowship**, Johns Hopkins University 2021 – 2022

- Awarded to a prospective CS PhD student who has shown exceptional promise; full tuition and stipend support

[5] **Howard and Jacqueline Chertkof Endowed Fellowship**, Johns Hopkins University 2021 – 2022

- Whiting School of Engineering donor-funded award supporting graduate financial aid

[4] **Science, Technology and Research Scholars (STARS) II Program**, Yale College 2019 – 2021

- Support for research, mentorship, & professional development workshops for underrepresented minority students

[3] **Alan S. Tetelman 1958 Fellowship for International Research in the Sciences**, Yale College 2019

- Provides support for original undergraduate research projects abroad in the natural and applied sciences

[2] **Science, Technology and Research Scholars (STARS) I Summer Program**, Yale College 2018

- Supports 1st or 2nd-year underrepresented minorities in STEM research, scientific communication development

[1] **Science, Technology and Research Scholars (STARS) I Academic Year Program**, Yale College 2017 – 2018

- Supports first-year underrepresented minorities in STEM research, workshops and a peer mentorship program

PUBLICATIONS

Peer-Reviewed Conference Papers

[C-3]. **K. T. Pineda,** E. Brown, & C. M. Huang. *"See You Later, Alligator": Impacts of Robot Small Talk on Task, Rapport, and Interaction Dynamics in Human-Robot Collaboration.*
Proceedings of 2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI) (2025)
Best paper award honorable mention | Top 5.25% of submissions

[C-2]. N. Salomons, **K. T. Pineda**, A. Adéjàre, & B. Scassellati. *"We Make a Great Team!": Adults with Low Prior Domain Knowledge Learn more from a Peer Robot than a Tutor Robot.*
Proceedings of 2022 ACM/IEEE International Conference on Human-Robot Interaction (HRI) (2022)

[C-1]. N. Tsoi, J. Connolly, E. Adéníran, A. Hansen, **K. T. Pineda**, T. Adamson, S. Thompson, R. Ramnauth, M. Vázquez, & B. Scassellati. *Challenges Deploying Robots During a Pandemic: An Effort to Fight Social Isolation Among Children.*
Proceedings of 2021 ACM/IEEE International Conference on Human-Robot Interaction (HRI) (2021)

Peer-Reviewed Journal Articles

[J-1]. G. Ajaykumar, **K. T. Pineda,** & C. M. Huang. (2023). *Older adults' expectations, experiences, and preferences in programming physical robot assistance.* International Journal of Human-Computer Studies, 180, 103127.

PREPRINTS

[M-1]. **K. T. Pineda,** A. Mahmood, & C. M. Huang. *"You Might Like It": How People Respond to Small Talk in Human-Robot Collaboration.* In: *arXiv preprint arXiv:2312.07454* (2023). [Under Review]

[M-2]. **K. T. Pineda,** B. Chien, A. Mishra, T. Williams, A. Guo, Z. Xiao & C. M. Huang. *Shaping Small Talk: Examining the Effects of Robot Disclosure in Collaborative Settings.* (2025). [Under Review]

TEACHING EXPERIENCE

Computer Science Head Teaching Assistant, Johns Hopkins University — Fall ’22, Fall ’23, Fall ’24, Spring’25
EN.601.490/690 Human-Computer Interaction

- Held weekly office hours, led in-class activities, wrote exams, graded materials, lectured on human-AI interaction

Lecturer in Computer Science, Yale University — Spring ’22
CPSC 470/570 Artificial Intelligence TF

- Held weekly office hours and attended staff meetings, graded coursework, gave course lecture & led exam review

Computer Science Learning Assistant, Yale University — Spring ’20
CPSC 223 Data Structures Undergraduate Learning Assistant (ULA)

- Held evening office hours, attended weekly staff meetings with instructor and assistants to prepare materials

Science and Quantitative Reasoning Tutoring Program, Yale University — Fall ’20
CPSC 223 Data Structures Peer Tutor

- Held one-on-one tutoring sessions with students to review course concepts and prepare for exams

STUDENT ADVISING & MENTORSHIP

Johns Hopkins University Master’s Students

Jiaming Du, Robotics — 10/2025 – Present
Kush Kumar Kushwaha, Robotics — 10/2025 – Present
Jie Wang, Computer Science — 10/2025 – Present
Brian Chien, Robotics | *Next: AI Intern at Hitachi* — 02/2025 – 08/2025

Johns Hopkins University Undergraduate Students

Anvii Mishra, Computer Science | ***2026 PURA Winner (14.47% acceptance rate)*** — 11/2024 – Present
Toluwani Williams, Electrical Engineering — 11/2024 – Present
Angela Guo, Computer Science — 09/2025 – Present
Ethan Brown, Neuroscience | ***Best paper award honorable mention*** | *Next: JHU HAMR Lab* — 01/2024 – 09/2024

SERVICE

Peer Reviewer for ACM/IEEE International Conference on Human-Robot Interaction — HRI ’24, ’25, ’26
Johns Hopkins LCSR Graduate Student Association, *President* — 01/2024 – Present

- Leader of an executive board managing $8,000/yr in student resources

Johns Hopkins Computer Science Graduate Student Council, *Social Committee* — 04/2022 – Present

- Organize bi-weekly social events for department members

Organizer for RSS 2022 Workshop — 01/2022 – 07/2022

- Close-Proximity Human-Robot Collaboration Workshop: Challenges and Opportunities

Yale Computer Science Departmental Student Advisory Committee (DSAC), *Board Member* — 01/2020 – 05/2020

- CS student representative to department faculty & administration, planned events for computer science students

PROFESSIONAL MEMBERSHIPS AND DEVELOPMENT

Cientifico Latino Graduate Student Mentorship Initiative (GSMI) — 08/2020 – 05/2021

- Mentorship program to help underrepresented students apply to graduate school through application preparation materials, one-on-one guidance, webinars, and mock-interviews

Cientifico Latino Graduate Student Engagement & Community (GSEC) — 08/2021 – 05/2022

- Mentorship program to help underrepresented students navigate their first year of graduate school through mentorship pods, one-on-one guidance, workshops, webinars, and socials

Yale Society of Women Engineers (SWE), *Vice President* — 08/2019 – 05/2020

- Organized professional development and community events for undergraduate women in engineering at Yale

PRESS AND MEDIA

Johns Hopkins University Whiting School of Engineering News –Making Robots More Human — 04/17/2025
Johns Hopkins University Whiting School of Engineering News –Celebrating 2025 department awardees — 05/09/2025

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, pandas), C, C++, C#, MATLAB, JavaScript, HTML, CSS, Linux, LaTeX, Git

Robotics-Hardware: Franka Emika Robot, Kinova Gen3, UR5, Fourier GR3, Arduino

Robotics-Software: ROS, Gazebo, RViz, MoveIt

Tools: Unity, AutoCAD, Adobe Illustrator, Figma, Pupil Labs Invisible (eye tracking)

Research: Human-Robot Interaction (HRI), Human-Computer Interaction (HCI), Human-Centered Design, Empirical Human-Subject Studies, Statistical Analysis (JMP, R), Qualitative Coding (MAXQDA)

AI Systems: Ollama, LangChain, Hugging Face, OpenAI API (ChatGPT), Speech-to-Text (Whisper, Google Cloud), Retrieval-Augmented Generation (RAG), Function Calling, Vector Databases (FAISS), Prompt Engineering, Local LLMs